

Instructor: Date:

Learner: Regional Branch:

Program: Center:

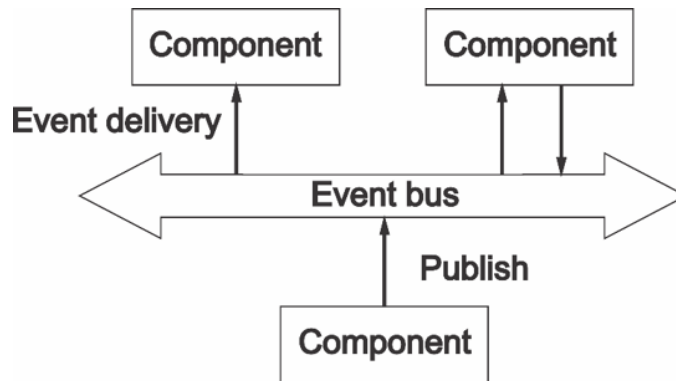
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Architecture Styles for Communication and Documentation



(a)

Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognize and implement the different types of architecture styles that can be included in the IT solution.

Ask your partner:

- What programs to store and share data do you know?
- What is the importance of debuggers in a system?

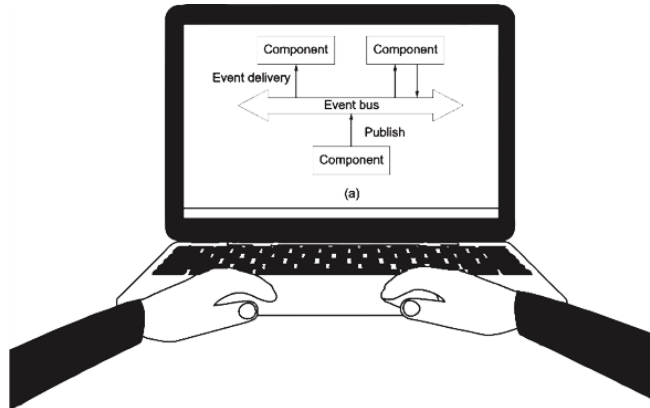
Can Do Statements:

- I can describe the event-based architecture style characteristics, advantages and disadvantages.
- I can talk about layered systems architecture style characteristics and advantages and disadvantages.
- I can give information about the repository architecture style.



Let's Explore!

Helping Michael study for his architecture styles test!



Instructions: Read the text below.

Michael: Help me out studying for my styles test! Take this list of questions.

Linda: What's the event-based architecture style about?

Michael: The event-based style is also known as implicit invocation. In this style, components invoke events instead of processes. Events operate the necessary processes implicitly to follow the component's instructions, like daemons. The characteristics of this style involve hooking up a particular process to an event, and also connectors that include procedure calls to bind such processes. Finally, in this style, we can find modules that provide a collection of procedures with a set of events.

Linda: What are its advantages and disadvantages?

Michael: Well, it's easily introduced into a system by including the events, it's a strong support for reuse, and it makes system evolution easier. On the other hand, components don't know how other components react and they relinquish control over computations performed in the system.

Linda: What's the layered systems architecture style about?

Michael: It's designed to work hierarchically among layers. Each layer sends and receives messages from layers above and below; this system is seen as a whole and it's able to provide detailed information about each layer. The characteristics involve clearness in the functionality of each layer and a constant stream of data flow.

Linda: Mention its advantages and disadvantages.

Michael: It's well-organized, layers have a defined task and they aren't dependent. Unfortunately, it's difficult to structure a layered system when it's intended to restrict communication to adjacent layers.

Linda: What's the repository architecture style?

Michael: It's a style composed by a central data structure and a group of components that manage the central data storage. It can be represented with a scheme of a blackboard architecture. It's used to back up information, manage a large amount of data, and share it among users.

**Let's Practice!**

Michael is checking out this information.

Instructions: Proofread the sentences. Correct the mistakes.

1. Michael has to figure in all the information about the repository architecture style.

2. A central data structure and a group of components make on a repository architecture style.

3. This style stands by with scheme of a blackboard architecture.

4. Engineers work in with this style because they can manage a large amount of data.

5. With this style many people can look up data and share them with others users.

**Let's Practice!**

These are the pros and cons of layered systems.

Instructions: Select the correct answers.

1. This system _____ as a whole and it's able to provide detailed information about each layer.
a. are seen
b. is saw
c. is seen
2. It's well-organized, layers _____ because they have a defined task.
a. are known
b. is know
c. are knew
3. This system _____ by many engineers because layers are independent.
a. are used
b. is used
c. is use
4. A layered system _____ by some people.
a. is hardly structure
b. are hardly structured
c. is hardly structured
5. The communication _____ to adjacent layers.
a. is restricted
b. are restricted
c. is restricting
6. This system _____ because it involves constant stream of data flow.
a. are uses
b. are used
c. is used





Extension Activity

Do you know these styles?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about event-based architecture, layered system architecture and repository architecture.
- Take notes about the information of each one system style.
- Write characteristics, advantages and disadvantages of each of those system styles.
- Make a brochure with the information that you have written in the chart.
- Share your brochure with your partners.

Self-Assessment

Yes

No

- I search information about event-based architecture, layered system architecture and repository architecture.
- I write characteristics, advantages and disadvantages of each system style
- I make a brochure with the information.
- I share with my partners.

☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- **daemon** (noun): computer program controlled by the user for background processes
e.g. Events operate the necessary processes implicitly to follow the component's instruction, like daemons.
- **implicitly** (adverb): something that is done without a direct command
e.g. Events operate the necessary processes implicitly to follow the component's instruction, like daemons.
- **scheme** (noun): a designed path to accomplish something
e.g. It can be represented with a scheme of a blackboard architecture.
- **relinquish** (verb): to desist from doing something
e.g. They relinquish control over computations performed in the system.
- **invoke** (verb): to summon something
e.g. Components invoke events instead of processes.



Learning Points

Phrasal Verbs

You use **phrasal verbs** to give a new meaning to a **verb** by adding a preposition or an adverb.

Phrasal verbs do not follow any specific rule; you just have to memorize them and learn their meanings. They are composed by a **verb** and a **preposition** or **adverb**.

There are some phrasal verbs that are separable, this means that you can separate the verb from the preposition or adverb when the action affects directly an object or object pronoun (*me, you, him, her, it, us, them.*)

Example:

Turn off

*The system is programmed to **turn off** the updates.*

*The system is programmed to **turn** the updated **off**.*

The system is programmed to **turn it off**. (Correct)

The system is programmed to ~~**turn off**~~ it. (Incorrect)

Some other phrasal verbs inseparable, so these words need to remain together, they cannot be separated.

Example:

Work on

*We are **working on** a new architecture style.*

There are thousands of phrasal verbs. Let's check some of them in the next chart:



Learning Points

Phrasal verb	Meaning	Example
Sign in	To enter into a web domain by using a username and password.	<i>I can't sign in on this page.</i>
Hook up	To connect something into something.	<i>Hook up the cable to the computer.</i>
Turn on	To power something so it starts.	<i>Turn the computer on.</i>
Boot up	The time that something takes to start.	<i>It takes a long time booting up this computer.</i>
Scroll up/down	To move a page up or down while navigating in it.	<i>Scroll down the page, I want to keep reading.</i>

Examples

- *Help me out studying for my styles test!*
- *The characteristics of this style involve hooking up a particular process to an event.*
- *We can find out modules that provide a collection of procedures with a set of events.*
- *It's used to back up information.*

Event-Based, Layered Systems, Repository Architecture Styles

Among the architecture styles, we find **event-based**, **layered** and **repository**. They were all designed with a collection of procedures and functions for their components to interact with each other.

Event-based Style

This system is based on actors, constraint satisfaction, daemons, and packet-switched networks. Whenever it is necessary, components announce events instead of invoking direct procedures due to the fact that each event brings implicitly all the essential procedures for the component.

A debugger, which is a program designed to clear errors, is an example of this style. Instead of announcing the procedures to take, it shows an event. Event-based programs provide a strong support for the system's evolution.





Learning Points

Layered Style

This style is organized hierarchically; the software functionality is divided into layers and these layers provide services to the one above and receive services from the one below. This style sees the system as a whole and is able to provide detailed information about each layer and its connections.

An example of this style is the Open System Interconnection (OSI), which is a network protocol.

Repository Style

This style is composed of two main components. The first one is a central data structure that represents the current state of the program, and the second one is represented as a group of independent components that manage the central data structure. This is an effective style to manage large amounts of data and share it. Some examples of this style are information systems, programming environments, or graphical editors.

Examples

- *The event-based style is also known as implicit invocation.*
- *It's designed to work hierarchically among layers. Each layer sends and receives messages from layers above and below.*
- *It's a style composed by a central data structure and a group of components that manage the central data storage.*